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Pharmacological Modification of Experimental Depression in Infant Macaques

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Abstract. This study provides evidence that mother–infant separation in macaques is a useful experimental model of depression. At the age of 6–8 months, seven *M. fascicularis* infants underwent two consecutive separations from their mothers lasting 21 and 15 days, respectively. The frequency and duration of a set of individual and social behaviours were recorded throughout each of the following experimental conditions: baseline, separation, reunion. In response to maternal separation, the infants showed marked increases in frequency of behaviours reflecting distress, self-directed activity or anxiety (e.g. vocalization, locomotion, body play). Both individual and social play behaviours were markedly suppressed in separated infants. During the second separation, one group of subjects was given, in a 'double blind' fashion, daily doses of 5 mg/kg of an antidepressant, desmethylimipramine (DMI), i.m. Treatment with DMI markedly diminished most of the behavioural alterations induced by separation. In particular, the increases in distress and self-directed behaviours as well as the suppression of play activities were prevented or antagonized. Plasma levels of DMI after 5 days of administration were in the range of 50–150 ng/ml.

Key words: Experimental depression – Desipramine – Monkeys – Separation

Mother–infant separation in monkeys has been proposed as a model of human anaclitic depression by Kaufman and Rosenblum (1967), after it was noticed that lower primate reaction to this stress was quite similar to that seen in humans (Burlingham and Freud, 1944; Spitz and Wolf, 1946, Bowlby et al., 1961). From the first primate separation experiments (Seay et al., 1962; Jensen and Tolman, 1962; Seay and Harlow, 1965; Hinde et al., 1966) to the recent ones (McKinney et al., 1973b; Hrdina and von Kulmiz, 1978), the procedure has consistently produced behavioural responses often labelled as 'protest' and 'despair'. The general pattern and the specific variabilities of the reaction have been described in detail (Lewis et al., 1976) and critically reviewed by Chappel and Meier (1975) and McKinney (1976).

An experimental model of depression should parallel as closely as possible depression in humans. Requirements for such a model, as suggested by McKinney (1976), include: (1) Stimuli controlling such behaviours should be similar to those controlling parallel behaviours in man. (2) The behaviours changed and produced should be similar to those seen in man. (3) The behaviour pattern should persist beyond the time of induction. (4) Treatment methods that alleviate signs of depression in man should have a similar effect in animal subjects. (5) The subjects must be capable of strong social bonds or 'affectional systems'.

The mother–infant separation paradigm in macaques was shown to fulfil most of these requirements. The purpose of the present study was to investigate whether administration of an antidepressant drug, desmethylimipramine, would influence any of the behavioural responses of infant macaques to maternal separation. We predicted that treatment of separated infants with this tricyclic antidepressant would produce behavioural changes in the direction opposite to those produced by separation; that is, in the direction of recovery.

Materials and Methods

Eight mother–infant pairs of crab-eating macaques *(M. fascicularis)* served as subjects for this experiment; one infant died, leaving seven to participate in the experiment. Two of the infants were male and five were female. Each infant had been born in the laboratory colony room to mothers known to be multiparous. The pairs lived together in their home cages since the birth of infants. At 4 months of age, they were introduced to a playpen apparatus similar

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to that used by Seay and Harlow (1965). Four primate living cages were attached to a specially constructed wire-mesh common playpen, which could be subdivided into four individual play units. A small door between each home cage and the playpen allowed the infants access to the play area but restricted the mothers to the living cage. The home cages were modified so that their occupants were directly observable (see Hrdina and von Kulmiz, 1978). The animals were allowed to become adapted to the apparatus for at least a month before baseline observations began. Food rations for the monkeys consisted of Purina monkey chow (without isoniazid), which was fed twice daily and was supplemented by fresh fruit and vegetables.

All subjects went through two separations lasting 21 and 15 days, respectively, and were subjected to isolation simultaneously. The separation was initiated with minimal distress by closing the door behind the infant as it entered the playpen. During the separation period, the infants lived alone in their home cages and were only allowed a limited period (half an hour) of social interaction a day. After each separation period, the mothers were returned to home cages and reunited with their infants.

Baseline measurements were made for 2 weeks before the first separation, at which time the mean age of the infants was 5.5 months. The conditions of the first separation were the same for all subjects. Before the second separation, one male infant was assigned to the control group and the other male to the drug treatment group; five infant females were assigned at random to the two groups. The drug-treated group ($n = 4$) received 5 mg/kg/day i.m. desmethylimipramine (DMI) for a week prior to and during 15 days of the separation period; the control group ($n = 3$) received an equal volume of physiological saline. The injections were administered in the evening after the last recording session in order to minimize the interference of injection stress with the behaviours measured. The drug phase of the experiment employed a double-blind procedure. Neither the observers nor the person administering the injections knew which subjects were receiving the drug and which were given saline.

The infants were tested in pairs during their free play time. The schedule of observation sessions was designed in a way which allowed for both the distribution of infant pairing and diurnal variation in activity levels. Owing to the death of one infant, pair assignments originally planned as a Latin square design for the purpose of counter-balanced observations had to be adjusted; each subject thus served as a social partner to another infant more frequently than would have been required had eight subjects been available. The individual behaviour of each animal was recorded for a 20-min period during the free play time each day for at least 4 days per week. Two observers worked simultaneously during the recording sessions, each observing one infant of the pair. They were trained to code as many as 32 ongoing behaviours by using a specially designed electronic recording apparatus that encodes both frequency and duration of behavioural events. Definitions of the behavioural items are given in the legend to Fig. 1. The final decoding of the recorded data was accomplished by means of a PDP 8E computer and teletype printer. Inter-observer reliability sessions of 5 min duration were taken at random during the experiment. The agreement for these sessions varied from 75–95%.

Data presented on 8 observation days during the 2 weeks of the baseline period, 13 observation days during 3 weeks of the first separation, and 7 observation days during the 13-day-long reunion period.

Data for the drug phase of the experiment are derived from 4 observation days during a week of the baseline condition prior to the second separation and from 10 observation days during the second separation. The mean frequency and duration of behaviours during each session was first calculated for individual infants and then averaged for all subjects in a group. The significance of differences between mean values for intragroup comparisons in different conditions was calculated by using the paired t-test whereas for intergroup comparison between control and experimental group a two-tailed Student's t-test was used. The difference was considered statistically significant for $P < 0.05$. In addition, analysis of variance of the original scores from control and experimental groups was done for observation periods before the second separation (BL_2) and during the second separation (S_2).

Plasma levels of DMI were measured by the external labelling technique of Hammer and Brodie (1967) as described in an earlier communication (Riva et al., 1975).

Results

The behavioural changes shown by the infants in response to maternal separation were similar to those reported in earlier separation studies. Immediately after separation, the characteristic signs of distress, such as increased vocalization and locomotion, were prevalent. These signs of protest persisted, although to a lesser extent, during the following few days. The next phase of separation, between the 4th and 7th days, was characterized by suppression of social and play behaviours, and increased passivity or what resembled despondency. A typical hunched-over posture considered by some workers as indicative of 'depression' was observed in some infants during this period.

Behavioural parameters showing marked changes in response to separation are reported in Fig. 1. The mean frequency of behaviours reflecting initial distress ('vocalization'), self-directed activity ('body play', 'eat and chew'), and passivity ('sit' and 'passive contact')

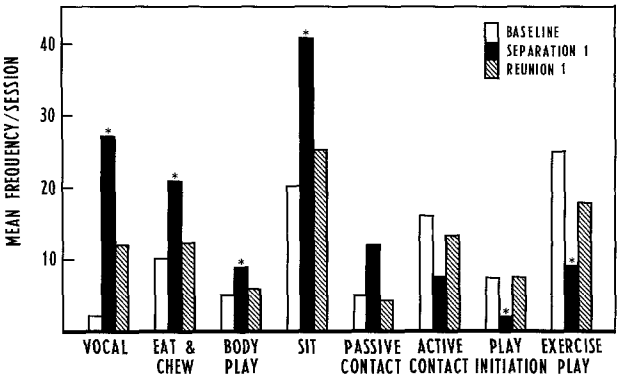


Fig. 1. Changes in several categories of behaviour in infant macaques as a consequence of maternal separation. Columns denote mean frequencies/session of individual behaviours in each of seven infants during 2 weeks of baseline observation (□), 3 weeks of separation (■) and 2 weeks of reunion (▨). Categories of behaviour were: *vocalization* (sounds of distress), *eat and chew* (chewing any object small enough to place in the mouth), *body play* (manipulation of the subject's own body, autogrooming), *sit* (sitting alone on haunches), *passive contact* (nonspecific passive physical contact), *active contact* (active non-specific physical contact initiated by the subject), *play initiating* (inducing another subject to play), *exercise play* (vigorous playful activity alone), and *social exploration* (manual, oral-olfactory exploration of a social partner). * Denotes significant difference (0.05) from baseline

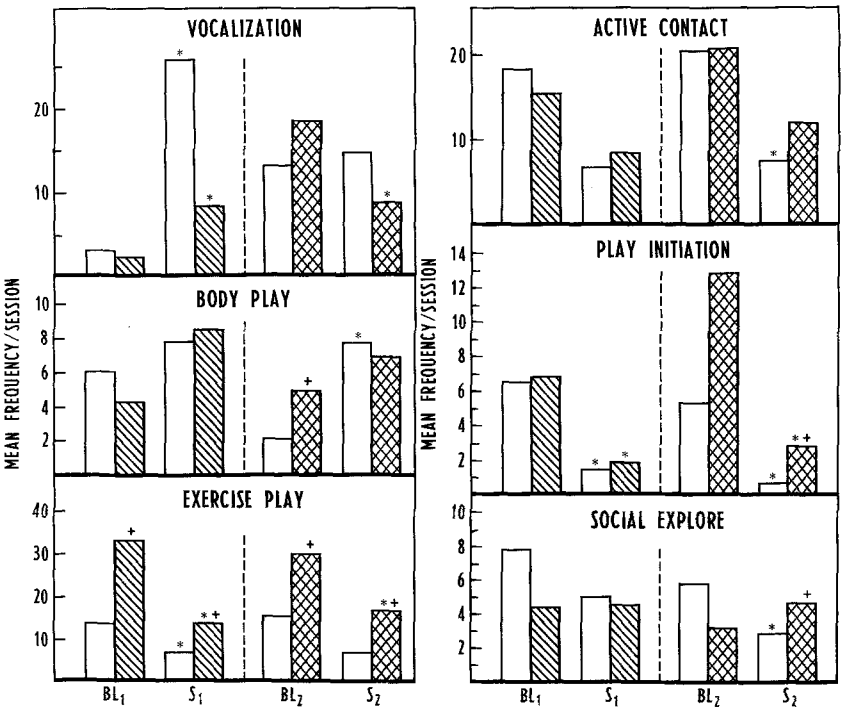


Fig. 2. Effect of desmethylimipramine (DMI, 5 mg/kg/day) treatment on behavioural responses of infant macaques to maternal separation. Columns represent mean values for frequency/session of various behaviours in three control (□) and four experimental (▨) subjects during the baseline observation period before the first separation (BL_1), during the first separation (S_1), during the baseline observation period before second separation (BL_2), and during the second separation (S_2). No treatment was administered before and during the first separation (BL_1 and S_1). Experimental subjects received DMI treatment prior to and during the second separation (BL_2 and S_2). * Significantly different from pre-separation value ($P < 0.05$). $^+$ Significantly different from control group in the same condition ($P < 0.05$)

was significantly (except for 'passive contact') higher during the separation period than during the pre-separation baseline. In contrast, the frequency of both individual ('exercise play') and social play behaviours ('active contact', 'play initiation') showed significant suppression in separated infants (Figs. 1 and 2). Upon reunion with the mother, frequency of most behaviours returned into the range of pre-separation values (Fig. 1).

Behavioural responses of infants to the second separation were qualitatively similar to those seen during the first separation. Treatment with DMI resulted in modification of some responses to separation, particularly those of the social behavioural type. The significant suppression of the frequency (Fig. 2) and duration (Fig. 3) of 'active contact' was prevented by the antidepressant drug. The frequency of 'play initiation' during the separation was significantly less suppressed in DMI-treated infants (Fig. 2). In response to separation, the frequency of 'social explore' was significantly decreased in controls, but not in experimental animals receiving DMI. The separation-induced increase in frequency of 'body play' was reduced in infants given the antidepressant when compared to that in saline-injected controls. In con-

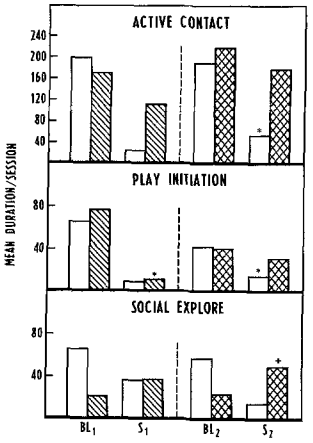


Fig. 3. Effect of desmethylimipramine (DMI, 5 mg/kg/day) treatment on behavioural responses of infant macaques to maternal separation. Columns represent mean values for duration (0.4-s scans/session) of various behaviours in three control (□) and four experimental (▨) subjects during the baseline observation period before the first separation (BL_1), during the first separation S_1), during the baseline observation period before the second separation (BL_2), and during the second separation (S_2). No treatment was administered before and during the first separation (BL_1 and S_1). Experimental subjects received DMI treatment prior to and during the second separation (BL_2 and S_2). * Significantly different from pre-separation values ($P < 0.05$). $^+$ Significantly different from control group in the same condition ($P < 0.05$)

trast, treatment with the antidepressant drug did not produce any significant effect on the suppression of 'exercise play', which is one of the typical responses of infants to separation. The frequency of vocalization which increased significantly in response to the first separation in both the control and the experimental groups was significantly suppressed in response to the second separation in the DMI-treated group when compared to pre-separation (BL_2) values. Analysis of variance has revealed that during the baseline observation period before the second separation (BL_2), there was no significant difference in variance between data from control and experimental animals for various behaviours except for 'body play' ($F = 4.53$, $d.f. = 1/26$, $P < 0.05$) and 'exercise play' ($F = 13.12$, $d.f. = 1/26$, $P < 0.01$). During the second separation (S_2), significant differences between the two groups were obtained for frequency of 'vocalization' ($F = 4.77$, $d.f. = 1/56, P < 0.05$), 'play initiation' ($F = 4.77$, $d.f = 1/56$, $P < 0.05$) and 'social explore' ($F = 5.49$, $d.f. = 1/56$, $P < 0.05$) as well as for duration of 'social explore' ($F = 6.54$, $d.f. = 1/56$, $P < 0.05$). Of behaviours showing significant intergroup variation during BL_2, 'exercise play' continued to be significantly different ($F = 15.41$, $d.f. = 1/56$, $P < 0.01$) whereas 'body play' did not ($F = 0.039$, $d.f. = 1/56$, N.S.).

Preliminary experiments in another group of three infant macaques have shown that administration of 5 mg DMI/kg per day for 5 days resulted in drug plasma levels averaging 149 $\pm$ 32 ng/ml at 4 h and 47 $\pm$ 10 ng/ml at 22 h after the last dose.

Discussion

Separation of infant macaques from their mothers caused reproducible behavioural changes characterized by reduction of active social contact and play behaviours, and a concomitant rise in behaviours indicative of social disengagement and passivity. This finding agrees with previous studies on mother–infant separation in monkeys (Kaufman and Rosenblum, 1967; McKinney et al., 1973b) in that typical signs of both protest and despair occurred in response to separation. Upon reunion, not all behaviours returned to pre-separation values, as exemplified by 'vocalization' and 'exercise play' (Fig. 1). This may be due, at least in part, to differences in maternal caretaking and reciprocity.

Treatment of infant macaques with a specific antidepressant drug, desmethylimipramine (DMI), reduced or even reversed some of the behavioural changes produced by separation. The effects of DMI were detected in categories of behaviour most affected by the separation procedure ('active contact', 'play initiation', 'social explore', and 'body play'). During separation, the DMI-treated infants showed a higher frequency than control animals of behaviours indicating active social participation. On the other hand, incidence of distress ('vocalization') and self-oriented behaviours ('body play') was suppressed in subjects receiving the antidepressant drug. DMI appears to act as a behaviour 'normalizer' to lessen the separation-induced behavioural responses.

During the baseline conditions, most behaviours did not differ significantly between the control and experimental group. Furthermore, modifications produced by DMI were in most cases quantitatively or qualitatively distinct from differences existing between the two groups during the second baseline period (BL_2). However, as can be seen in Fig. 2, significant differences, in the predicted direction, did exist during BL_2 in two behaviours ('body play' and 'exercise play'). One of these differences ('exercise play') was present even during the first baseline period (BL_1). It is conceivable that the treatment group may have been, by chance, more resistant to the separation stress; this then could have contributed to the apparent drug effect on some behaviours.

The dose schedule of DMI used in our study was likely to produce, as shown in our preliminary experiments, plasma levels that are within the range of those usually attained in depressed humans during effective therapy with tricyclics (Alexanderson, 1972; Åsberg, 1976).

The effect of drugs on isolation-induced, disturbed behaviours in primates has been reported previously. Whereas trifluoperazine, dextroamphetamine sulphate, and oxazepam were found to be ineffective in modifying the behavioural deficits in socially isolated chimpanzees (Menzel et al., 1963; Turner et al., 1969), administration of chlorpromazine to rhesus monkeys subjected to partial social isolation resulted in a significant decrease in their self-disturbance behaviour (McKinney et al., 1973a). In the latter study, however, chlorpromazine failed to increase the frequency of active social encounters in most of the subjects.

Recently, Suomi et al. (1978) reported that imipramine treatment yielded some behavioural improvement in young rhesus monkeys subjected to repetitive peer-separation. The demonstration that a clinically effective antidepressant drug differentially modified behavioural responses of infant macaques to maternal or peer-separation lends new support to the separation paradigm as a valid experimental model of depression. On the other hand, the effects seen with DMI may not be specific and could also be achieved with other drugs, e.g., anti-anxiety or antipsychotic. Future studies comparing the effect of chlorpromazine, diazepam, and another tricyclic antidepressant would help not only to

clarify this question but also to evaluate further the significance of separation-induced behavioural changes for studies of human psychopathology.

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